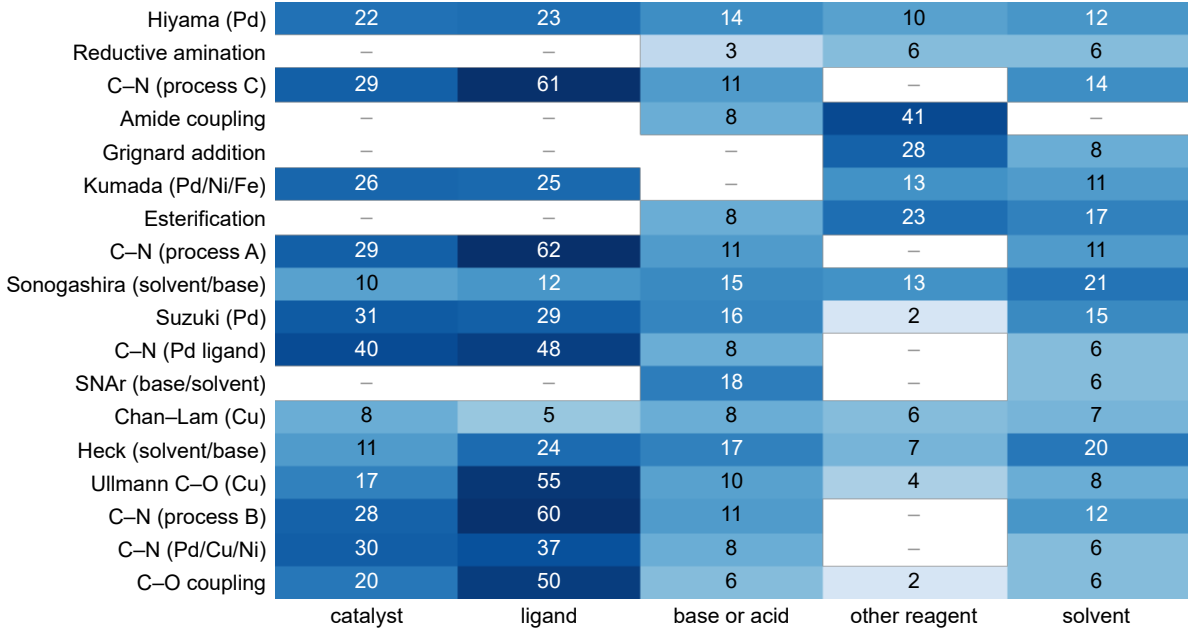
Hiyama (Pd) 22 23 14 10 12
Reductive amination – – 3 6 6
C–N (process C) 29 61 11 – 14
Amide coupling – – 8 41 –
Grignard addition – – – 28 8
Kumada (Pd/Ni/Fe) 26 25 – 13 11
Esterification – – 8 23 17
C–N (process A) 29 62 11 – 11
Sonogashira (solvent/base) 10 12 15 13 21
Suzuki (Pd) 31 29 16 2 15
C–N (Pd ligand) 40 48 8 – 6
SNAr (base/solvent) – – 18 – 6
Chan–Lam (Cu) 8 5 8 6 7
Heck (solvent/base) 11 24 17 7 20
Ullmann C–O (Cu) 17 55 10 4 8
C–N (process B) 28 60 11 – 12
C–N (Pd/Cu/Ni) 30 37 8 – 6
C–O coupling 20 50 6 2 6
catalyst
ligand
base or acid
other reagent
solvent